Project 1: Automate Docker built and push using Jenkins file.

1. Setup a Simple Flask App

Project Structure : app.py has main Flask Application File.

Code Blame 10 lines (7 loc) · 162 Bytes Raw

```python
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello_world():
    return 'Hello, World!'

if __name__ == '__main__':
    app.run(debug=True)
```

2. Docker File: Defines the Docker image for the Flask App.

Code Blame 20 lines (14 loc) · 518 Bytes Raw

```
# Use an official Python runtime as a parent image
FROM python:3.9-slim

# Set the working directory in the container
WORKDIR /app

# Copy the current directory contents into the container at /app
COPY . /app

# Install any needed dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Make port 5000 available to the world outside this container
EXPOSE 5000

# Define environment variable to avoid Python buffering
ENV PYTHONUNBUFFERED 1

# Run app.py when the container launches
CMD ["python", "app.py"]
```

3. Jenkinsfile : Contains the Jenkins Pipeline Configuration.

Code Blame 29 lines (25 loc) · 709 Bytes Raw

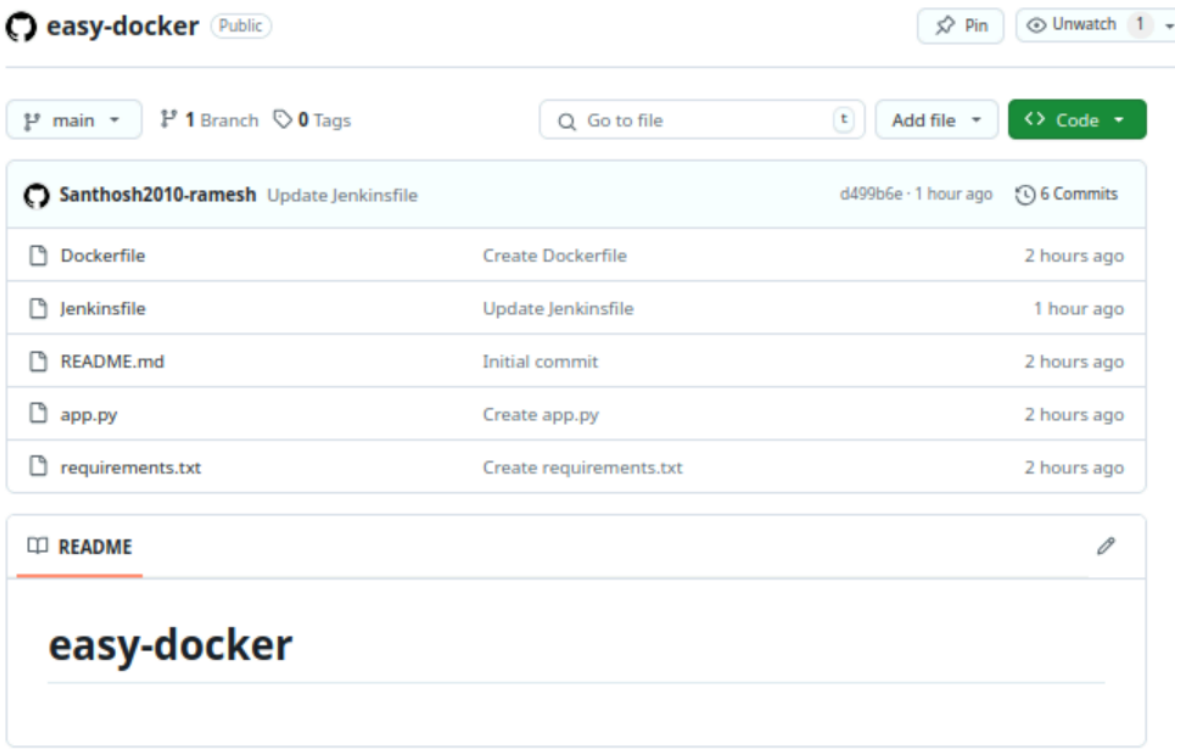
```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = 'santhosh2010/my-flask-app:latest'
    }

    stages {
        stage('Clone Repository') {
            steps {
                git url:'https://github.com/Santhosh2010-ramesh/easy-docker.git',branch: 'main'
            }
        }

        stage('Build Docker Image') {
            steps {
                sh 'docker build -t $DOCKER_IMAGE .'
            }
        }

        stage('Push Docker Image') {
            steps {
                withDockerRegistry([credentialsId: 'docker-hub-credentials', url: 'https://index.docker.io/v1/']) {
                    sh 'docker push $DOCKER_IMAGE'
                }
            }
        }
    }
}
```

4. **Requirements.txt List of Dependencies (Flask and other)**

5. **Push the code to GitHub**

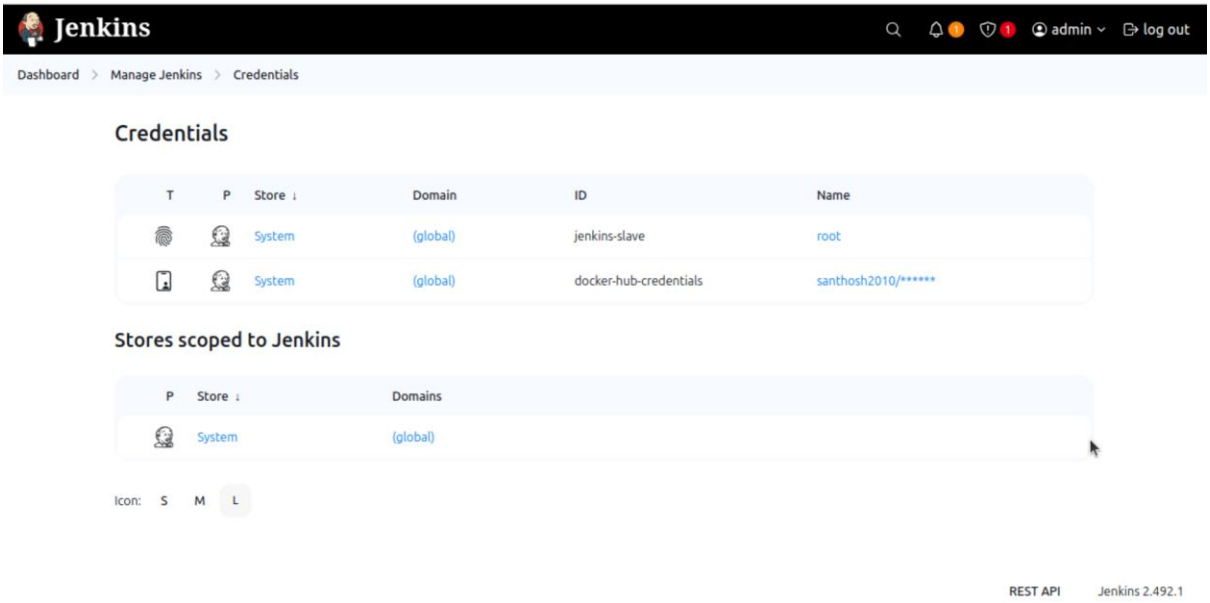
Make sure you have a GitHub repository created for the project.

Push all the files (app.py, requirements.txt, Docker File, Jenkins File) to the GitHub repository.

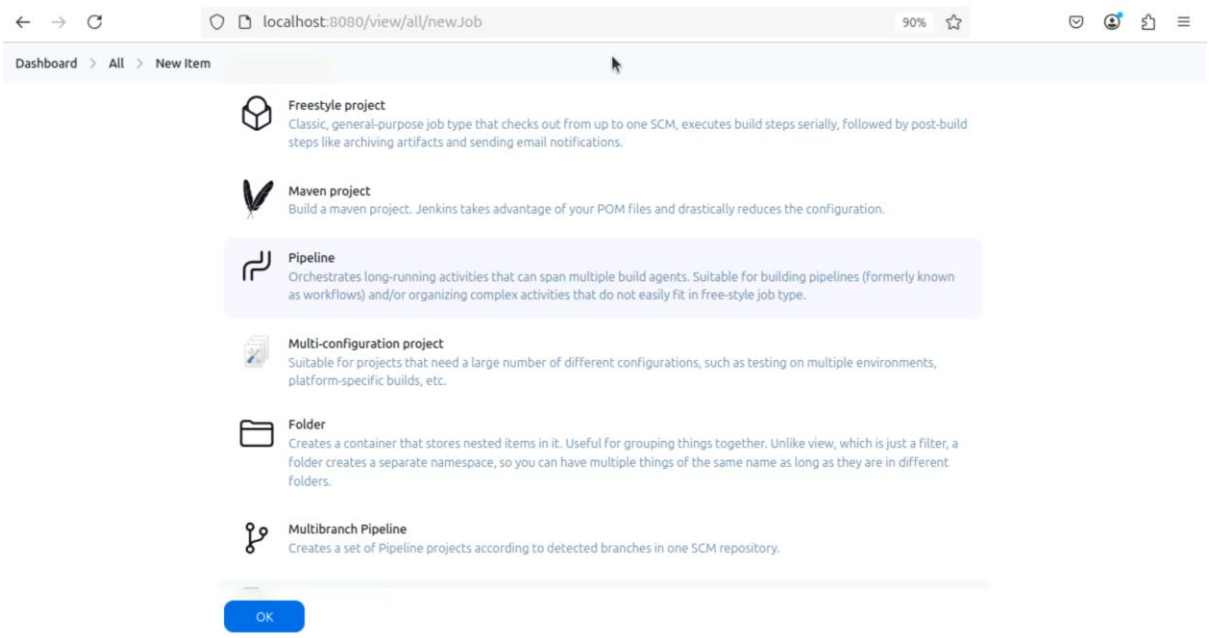


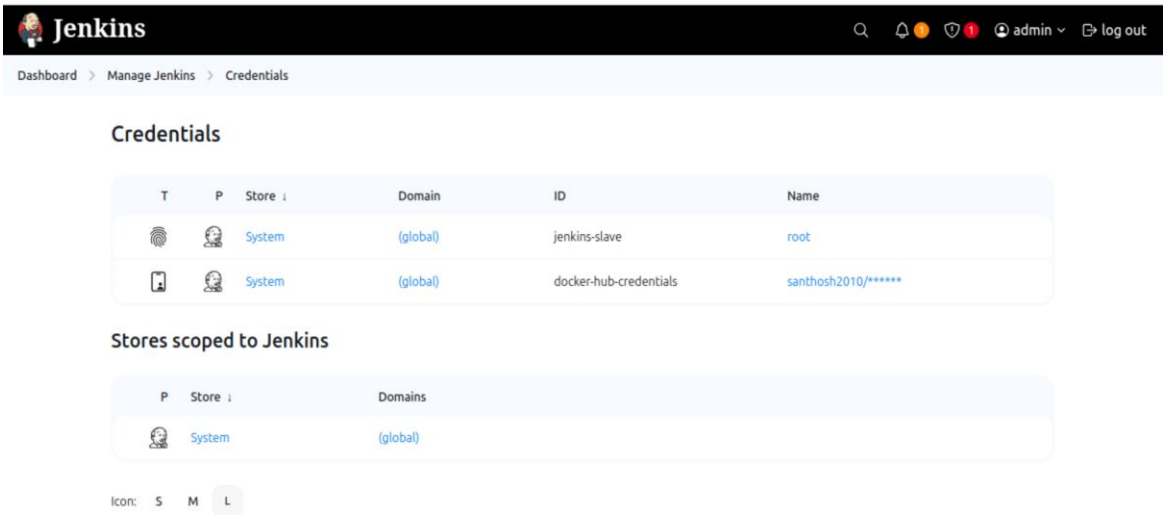
6. **Configure Docker Hub Credentials in Jenkins**
   Goto Jenkins > manage Credentials.

   Add new Credentials:
   Username & Password: You Docker Hub username and password.
   ID: Name it something like dockerhub-creds(the same name used in the Jenkinsfile)



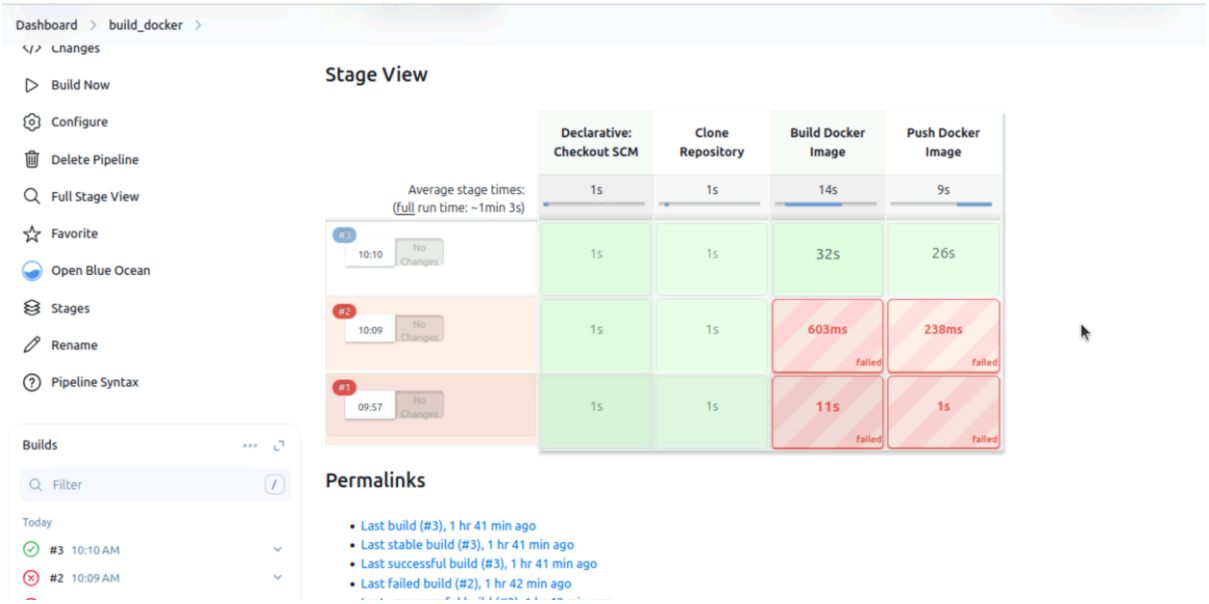
7. **Create a New Pipeline in Jenkinsfile**
   In Jenkinsfile, click New Item > Pipeline.
   Enter a new for the Pipeline.
   Under Pipeline definition, Select Pipeline Script from SCM.
   Select Git as the SCm.
   Enter the Github repository URL.
   Set the branch (typically master or main)
   Click Save

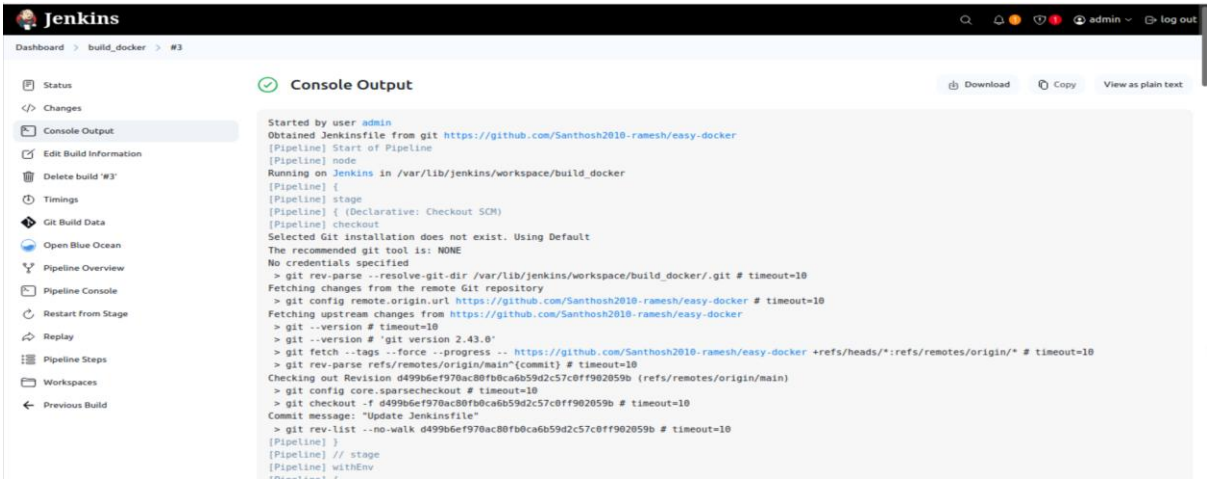




8. Click Build Now

Click Build Now in Jenkins to trigger the build.

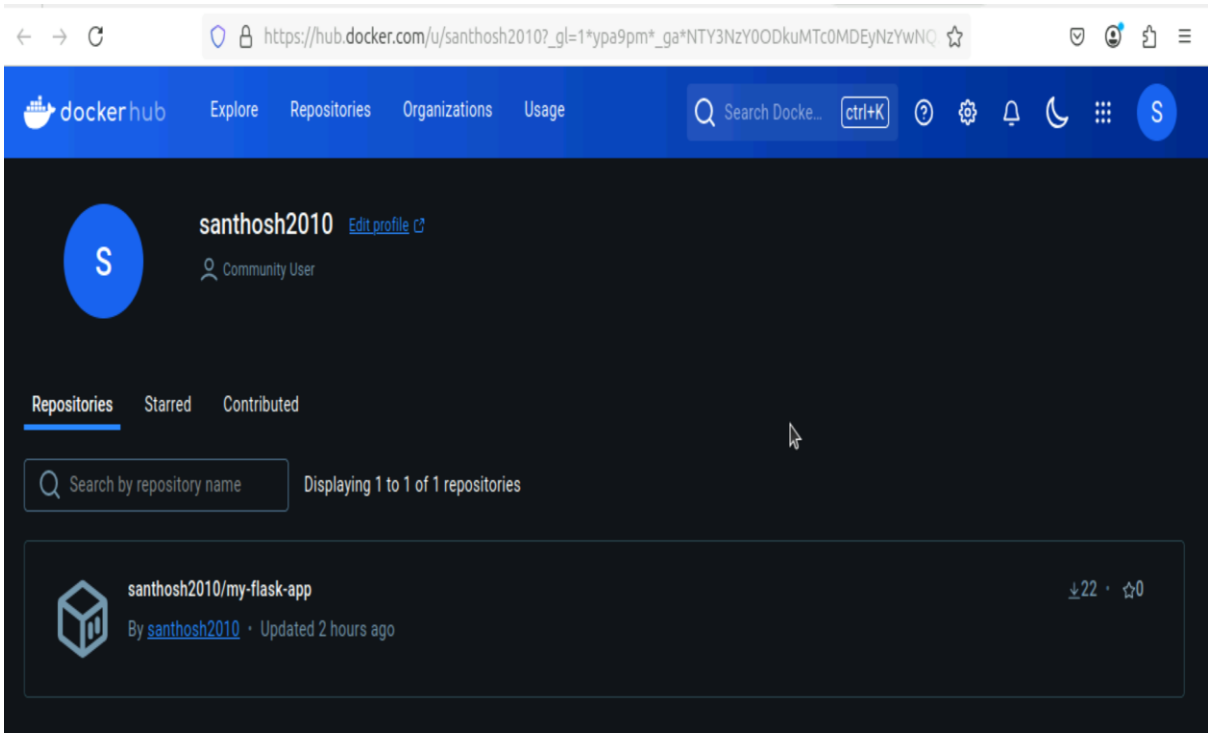




9. Verify Docker Image on Docker Hub.

After the Build Finishes, log into your Docker Hub Account.

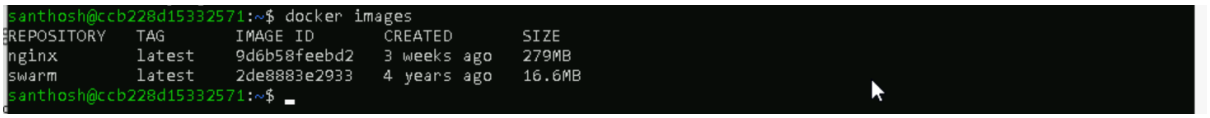
You should see the my_flask_app image under Repositories with the latest tag.



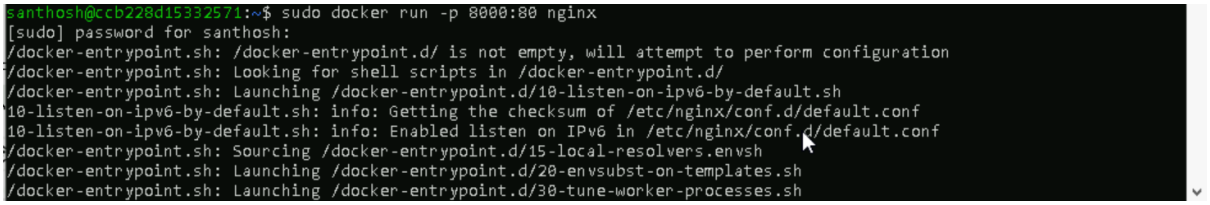
Project 2: Installing and enabling Docker Inside Ubuntu Terminal.

Step 1: Installing Nginx.

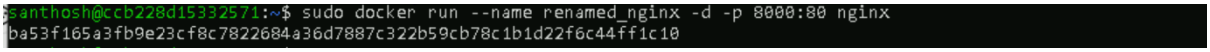
Step 2: Displaying the Image List.



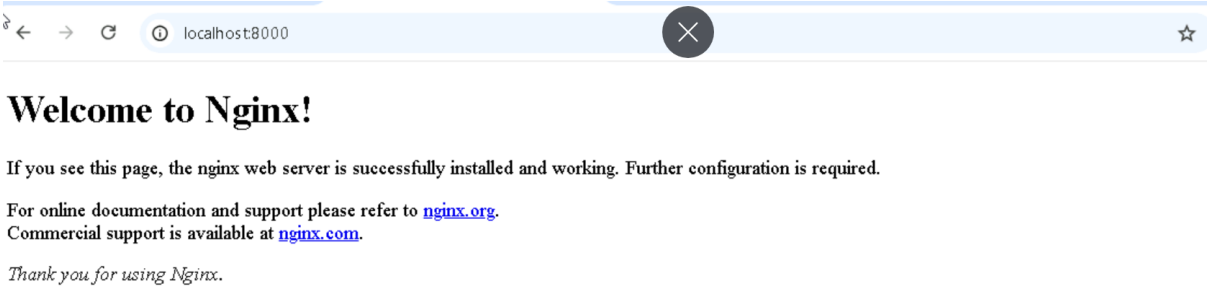
Step 3: Running the Container.



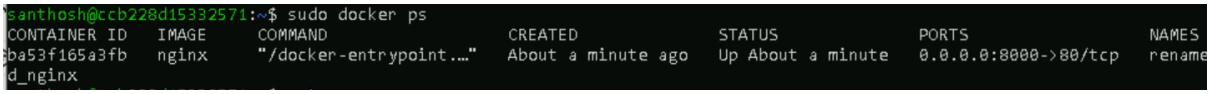
Step 4: Running the Renamed Container.



Step 5: Localhost:8000



Step 6: Displaying the running container.



Step 7: Interacting Directly with renamed Container.

```
santhosh@ccb228d15332571:~$ sudo docker exec -it renamed_nginx /bin/bash
root@ba53f165a3fb:/# _
```

Step 8: Modifying the index.html file.

```
The following NEW packages will be installed:
  libgpm2 libncursesw6 nano
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 838 kB of archives.
After this operation, 3339 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bookworm/main amd64 libncursesw6 amd64 6.4-4 [134 kB]
Get:2 http://deb.debian.org/debian bookworm/main amd64 nano amd64 7.2-1+deb12u1 [690 kB]
Get:3 http://deb.debian.org/debian bookworm/main amd64 libgpm2 amd64 1.20.7-10+b1 [14.2 kB]
Fetched 838 kB in 0s (6067 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libncursesw6:amd64.
(Reading database ... 7580 files and directories currently installed.)
Preparing to unpack .../libncursesw6_6.4-4_amd64.deb ...
Unpacking libncursesw6:amd64 (6.4-4) ...
Selecting previously unselected package nano.
Preparing to unpack .../nano_7.2-1+deb12u1_amd64.deb ...
Unpacking nano (7.2-1+deb12u1) ...
Selecting previously unselected package libgpm2:amd64.
Preparing to unpack .../libgpm2_1.20.7-10+b1_amd64.deb ...
Unpacking libgpm2:amd64 (1.20.7-10+b1) ...
Setting up libgpm2:amd64 (1.20.7-10+b1) ...
Setting up libncursesw6:amd64 (6.4-4) ...
Setting up nano (7.2-1+deb12u1) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/editor.1.gz because associated file /usr/share/man/man1/nano.1.gz (of link group editor) d
oesn't exist
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/pico.1.gz because associated file /usr/share/man/man1/nano.1.gz (of link group pico) doesn
't exist
Processing triggers for libc-bin (2.36-9+deb12u9) ...
root@ba53f165a3fb:/# nano /usr/share/nginx/html/index.html
```

Step 9: Editing the line on Nginx Html file.

```
santhosh@ccb228d15332571: ~
  GNU nano 7.2                                  /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Step 10: Nginx Localhost 8000 site.